

Chapter-3: Vector Differentiation

Q. Vector Differentiation

Q. Space curve

Q-6. Find the unit tangent vector at the point where $t = 2$ on the curve $x = t^2 + 1$, $y = 4t - 3$, $z = 2t^2 - t$.

Exercise- 33

Q. Find the angle between the tangent vectors of the curve $x = t^2 + 1$, $y = 4t - 3$, $z = 2t^2 - t$ at the point where $t = \pm 2$.

Q. Velocity and acceleration

Q-4. A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$, where t is the time. Find the components of its velocity and acceleration at time $t = 1$ in the direction $\hat{i} - 3\hat{j} + 2\hat{k}$.

Exercise- 3, 32

Q-9. If $\vec{A}$ has constant magnitude, then show that $\vec{A}$ and $\frac{d\vec{A}}{dt}$ are perpendicular provided $\frac{d\vec{A}}{dt} \neq 0$.

Q-18. Prove the Frenet-Serret formula:-

Q-19. Given a space curve $x = 3 \cos t$, $y = 3 \sin t$, $z = 4t$ and find (a) the unit tangent $\vec{T}$, (b) the principal normal $\vec{N}$, (c) the binormal $\vec{B}$, (d) the curvature k , (e) the radius of curvature ρ , (f) the torsion τ , (g) the radius of torsion σ .

Exercise- 22, 47-60

Q-22. Given the space curve $x = t, y = t^2, z = \frac{2t^3}{3}$, find (a) the curvature k (b) the torsion τ .

Q-23. Find the equations in vector and rectangular form for the (a) tangent, (b) principal normal, and (c) the binormal to the curve of Problem 22 at the point where $t = 1$.

Exercise- 60

Q-24. Find the equations in vector and rectangular form for the (a) osculating plane, (b) normal plane, and (c) rectifying plane to the curve of Problem 22 and 23 at the point where $t = 1$.

Exercise- 61

Q-25. (a) Show that the equation $\vec{r} = \vec{r}(u, v)$ represents a surface.

(b) Show that the $\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v}$ represents a vector normal to the surface.

(c) Determine a unit normal to the following surface, where $a > 0$:

$$\vec{r} = a \cos u \sin v \hat{i} + a \sin u \sin v \hat{j} + a \cos v \hat{k}$$

Q-26. Find the equation of the tangent plane to the surface $z = x^2 + y^2$ at the point $(1, -1, 2)$.

Exercise- 63, 64

Q-27. Show that the acceleration $\vec{a}$ of a particle which travels along a space curve with velocity $\vec{v}$ is given by

$$\vec{a} = \frac{dv}{dt}\vec{T} + \frac{v^2}{\rho}\vec{N}$$

where $\vec{T}$ is the unit tangent vector to the space curve, $\vec{N}$ is its unit principal normal, and ρ is the radius of curvature.

Q-66. A particle moves along the curve $x = (t^3 - 4t)$, $y = (t^2 + 4t)$, $z = (8t^2 - 3t^3)$, where t is the time. Find the magnitudes of the tangential and normal components of its acceleration at time $t = 2$.